

Research Methodology in CS (CSEG3060_3)

Lecture 2: Overview of Patents, Designs, and Copyrights: Definition, scope, duration, rights conferred, and key differences between Patents, Industrial Designs, and Copyrights under Indian and international IP frameworks.

Unit 5: Intellectual Property and Patents • Mapped Course Outcome: CO1

Lecture 2: Overview of Patents, Designs, and Copyrights

Unit 5: Intellectual Property and Patents

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science) **Semester:** V **Mapped Course Outcome:** CO1 **Estimated Duration:** 60 minutes

1. Learning Objectives

By the end of this lecture, students will be able to:

1. Define and distinguish between the three principal categories of Intellectual Property (IP): Patents, Industrial Designs, and Copyrights.
2. Describe the legal scope, subject matter, and eligibility criteria for each form of IP protection.
3. State the statutory duration and renewal provisions governing each IP right.
4. Identify the exclusive rights conferred upon the proprietor under the relevant statutes.
5. Compare and contrast Patents, Designs, and Copyrights across multiple legal parameters.
6. Locate these rights within the Indian statutory framework (Patents Act, 1970; Designs Act, 2000; Copyright Act, 1957) and the international IP architecture (Paris, Berne, TRIPS, PCT).

2. Introduction

Intellectual Property (IP) refers to creations of the mind—intellectual creations for which exclusive rights are recognized under the law. In computer science and engineering, IP is not a peripheral topic; it is foundational. The code you write, the algorithms you design, the user interfaces you conceptualize, and the datasets you curate all potentially qualify for one or more forms of IP protection.

This lecture dissects the three most relevant IP regimes for a computer science researcher: **Patents**, **Industrial Designs**, and **Copyrights**. Each protects a distinct category of creative output, has its own statutory foundation, and confers a different bundle of rights. A clear understanding of these three regimes is essential for researchers engaged in publishing, productizing innovations, or licensing technology.

The lecture is organized in four parts:

- Part A: Patents
- Part B: Industrial Designs
- Part C: Copyrights
- Part D: Comparative Analysis and International Framework

3. Part A: Patents

3.1 Definition

A **patent** is a statutory right granted by the government to an inventor, conferring upon the inventor the exclusive right to make, use, sell, and import the patented invention for a limited period, in exchange for public disclosure of the invention.

Statutory definition (Indian Patents Act, 1970):

- **Section 2(1)(j)** defines an "invention" as a "*new product or process involving an inventive step and capable of industrial application.*"
- **Section 2(1)(l)** defines a "patent" as "*the patent granted under the Act.*"

In essence, patentability requires the satisfaction of three cumulative conditions:

1. **Novelty (New)** – The invention must not have been anticipated by prior art anywhere in the world before the date of filing (or priority date).
2. **Inventive Step (Non-obvious)** – The invention must involve a feature that would not be obvious to a person skilled in the art (PSA).
3. **Industrial Application (Useful)** – The invention must be capable of being made or used in an industry.

3.2 Subject Matter and Scope

What can be patented (Section 3 and 4 of the Patents Act, 1970):

In the context of computer science, the following are patentable subject matter:

- A novel **algorithm** when embodied in a concrete technical application (e.g., a new image compression method that produces measurable technical effects).
- A **hardware architecture** or chip design.
- A **computer-implemented invention** that produces a technical effect or solves a technical problem.
- A **method of manufacturing** or industrial process.
- A new **chemical compound**, biotech product, or pharmaceutical formulation.

What is not patentable in India (Section 3):

- An invention that is frivolous or contrary to public order or morality.
- Discovery of a scientific principle, theory, or mathematical method (in the abstract).
- Schemes, rules, or methods of performing mental acts or playing games.
- Presentation of information.
- Topographies of integrated circuits (governed separately).
- A mere algorithm.
- Plants and animals in whole or part (with exceptions for microbiological processes).
- Traditional knowledge per se (unless combined with a technical application).
- An invention which is effectively traditional knowledge.

The boundary between a patent-eligible computer-implemented invention and a non-patentable abstract algorithm is one of the most actively litigated questions globally.

3.3 Duration

Under **Section 53 of the Patents Act, 1970**, the term of a patent in India is:

$$T_{\text{patent}} = 20 \text{ years from the date of filing the application}$$

This aligns with **Article 33 of the TRIPS Agreement**, which mandates a minimum protection period of 20 years from the filing date. There is no provision for extension of the term beyond 20 years.

3.4 Rights Conferred (Section 48)

Once a patent is granted, the patentee enjoys the following exclusive rights:

1. Right to **prevent** third parties from making the patented product.
2. Right to **prevent** third parties from using the patented process.
3. Right to **prevent** third parties from offering for sale, selling, or importing the patented product or the product obtained directly by the patented process.
4. Right to **assign** or **license** the patent to others (voluntary licensing).
5. Right to seek **relief in respect of infringement** (injunctions, damages, accounts of profits).

The patentee does not have a positive right to practice the invention; rather, they have a negative right to exclude others. If a complementary patent is held by another party, the patentee may be subject to a "blocking" situation and may seek a **compulsory license** under Section 84.

3.5 Patentable Computer Science Examples

Invention	Patentable Subject	Patent Number (Illustrative)
LZW compression method	Software-implemented process	US 4,558,302
PageRank algorithm	Claim tied to search engine method	US 6,285,999
RSA public-key cryptosystem	Cryptographic method	US 4,405,829
Method for distributed consensus (Paxos family)	Process for distributed computing	Multiple

4. Part B: Industrial Designs

4.1 Definition

An **industrial design** is a form of intellectual property that protects the visual appearance or aesthetic features of an article. It concerns how an article *looks*, not how it *works*.

Statutory definition (Designs Act, 2000, Section 2(d)):

"design means only the features of shape, configuration, pattern, ornament or composition of lines or colours applied to any article whether in two dimensional or three dimensional or in both forms, by any industrial process or means."

Three cumulative conditions govern registrability:

1. **Novelty** – The design must be new or original.
2. **Distinctiveness** – It must be significantly distinguishable from known designs.
3. **Not disclosed to the public** – It must not have been published or used in India or elsewhere before the filing date.

4.2 Subject Matter and Scope

What is registrable:

- The **shape** of a smartphone, a chair, or a computer peripheral.
- The **configuration** of a car's body panel.
- The **pattern** on a piece of fabric or wallpaper.
- The **ornament** or surface decoration of a consumer product.
- **Two-dimensional** designs (icons, logos applied to a product, graphical user interface elements) — *with limits*.
- **Three-dimensional** designs (the contour or form of an article).

What is not registrable (Section 4 of the Designs Act, 2000):

- Designs that are not new or original.
- Designs that have been disclosed to the public prior to filing.
- Designs that are contrary to public order or morality.
- A mere mechanical contrivance or principle.
- A mere workshop improvement or mode of construction.
- Trade marks or property marks.
- A design that is solely dictated by function (the "must-fit" exclusion).

4.3 Duration (Section 11)

The total protection period is:

$$T_{\text{design}} = 10 \text{ years (initial)} + 5 \text{ years (extension)} = 15 \text{ years maximum} \quad T_{\text{design}} = 10 \text{ years (initial)} + 5 \text{ years (extension)} = 15 \text{ years maximum}$$

The initial period is 10 years from the date of registration. An extension of 5 years may be sought by making an application in the prescribed form before the expiry of the initial period.

4.4 Rights Conferred (Section 22)

The registered proprietor of a design has the exclusive right to apply the design to any article in the class in which the design is registered. Specifically, the proprietor may:

1. Sue for **infringement** under Section 22, which incorporates the provisions of the Copyright Act regarding offences, penalties, and remedies.
2. **Assign** or **license** the design.
3. **Recover damages** for unauthorized application of the design.
4. Seek **injunctions** against further infringing acts.

In computer science, industrial designs are most relevant for:

- The physical form of computing devices (laptops, smartphones, wearables).
- The overall visual design of consumer electronics.
- Graphical User Interfaces (GUIs) — note that in India, GUI designs may be registered under the Designs Act as of 2008, although there are restrictions on purely functional or animated GUIs.

4.5 Key Distinction: Design versus Patent

A design protects **appearance**; a patent protects **function**. A novel laptop hinge mechanism may be patented, but the visual appearance of the laptop's casing is registrable as a design. Both can coexist on the same product.

5. Part C: Copyrights

5.1 Definition

Copyright is a bundle of legal rights that arises automatically upon the creation of an original literary, dramatic, musical, artistic, or other qualifying work. Unlike patents and designs, copyright does not require formal registration for subsistence (although registration under Section 44 of the Copyright Act, 1957 provides procedural advantages and is taken as prima facie evidence).

Statutory definition (Copyright Act, 1957, Section 13):

Copyright subsists in the following classes of works:

1. **Original literary works** – books, computer programs, software code, tables, compilations, databases (where original).
2. **Dramatic works** – scripts, choreographic works.
3. **Musical works** – compositions (with or without words).
4. **Artistic works** – paintings, sculptures, drawings, engravings, photographs, architectural works.
5. **Cinematograph films** – moving images with synchronized sound.
6. **Sound recordings** – recordings of sounds in any medium.
7. **Architectural works** – the artistic character and design of buildings.

5.2 Subject Matter and Scope

Copyright in Computer Programs (Section 2(o)):

Computer programs are explicitly treated as "literary works" under the Copyright Act, 1957. The legal protection extends to the **expression** of the program — that is, the source code, object code, and certain structural and organizational elements — but not to the **ideas, algorithms, methods of operation, or underlying logic** (Section 39 of the Copyright Act for technological protection measures and Section 66 for offences).

Examples in computer science:

- Source code in any programming language.
- Object code (compiled form).
- Documentation and user manuals.
- Website content (text, graphics, layout).
- Software architecture diagrams (as artistic works).
- Database contents (where there is originality in selection/arrangement).
- Multimedia content (a combination of multiple copyrighted works).

The Idea-Expression Dichotomy:

Copyright protects the expression of ideas, not the ideas themselves. This is the central tenet codified by Section 13 and reinforced by the Berne Convention. Therefore, the algorithm for sorting arrays is not copyrightable, but a specific implementation in C++ is.

5.3 Duration (Section 22 to 28)

The duration varies based on the nature of the work:

$$T_{copyright} = \begin{cases} \text{Life of the author} + 60 \text{ years} & \text{for literary, dramatic, musical, artistic works} \\ 60 \text{ years from the year of publication} & \text{for cinematograph films, sound recordings, photographs, government works} \\ 60 \text{ years from the year of publication} & \text{for anonymous/pseudonymous works and works of joint authorship} \end{cases}$$

Life of the author + 60 years

60 years from the year of publication

60 years from the year of publication

for literary, dramatic, musical, artistic works

for cinematograph films, sound recordings, photographs, government works

for anonymous/pseudonymous works

Tcopyright = [,]

The term is calculated from the calendar year next following the death of the author (for natural person authors) or from the year of first publication (for corporate and statutory works).

5.4 Rights Conferred (Sections 14, 57)

Copyright confers two distinct sets of rights:

A. Economic Rights (Section 14):

The owner has the exclusive right to:

1. Reproduce the work.
2. Issue copies to the public.
3. Perform the work in public.
4. Communicate the work to the public.
5. Make translations, adaptations, or arrangements.
6. Make cinematograph films or sound recordings of the work.

B. Moral Rights (Section 57):

The author retains, even after assignment of economic rights:

1. **Right of paternity** – the right to claim authorship of the work.
2. **Right of integrity** – the right to restrain distortion, mutilation, or modification of the work that would be prejudicial to the author's honour or reputation.
3. **Right of retraction** – in certain circumstances, to withdraw the work from circulation (subject to compensation to the assignee).

Moral rights are perpetual, inalienable, and non-assignable.

5.5 Copyright versus Software Patents

This is a critical distinction for computer science researchers:

Aspect	Copyright in Software	Software Patent
Protects	Expression (code, structure)	Underlying functional concept
Requires registration	No (automatic)	Yes
Duration	Life + 60 years	20 years
Disclosure requirement	No	Yes (full disclosure)
Cost of acquisition	Minimal	Substantial

In practice, software is typically protected by both copyright (on the code) and, where applicable, patents (on the underlying method).

6. Comparative Analysis: Patents vs Designs vs Copyrights

The following table summarizes the key differences across all three regimes:

Parameter	Patent	Industrial Design	Copyright
What is protected	New functional invention	Visual appearance of an article	Original literary/artistic expression
Governing statute (India)	Patents Act, 1970	Designs Act, 2000	Copyright Act, 1957
Nature of right	Negative right to exclude	Negative right to exclude	Bundle of economic + moral rights
Registration required	Yes (mandatory)	Yes (mandatory)	No (automatic upon creation)
Examination	Substantive examination	Formal examination only	Registration is optional; no examination
Duration	20 years from filing	10 + 5 = 15 years	Life + 60 years (typical)
Renewable	No	One extension of 5 years	N/A (extends automatically)
Novelty required	Absolute novelty worldwide	New and original	Originality of expression
Disclosure required	Full disclosure in specification	By representation in drawings	Not required
International convention	PCT, Paris Convention	Hague Agreement, Paris Convention	Berne Convention, TRIPS
Typical CS example	Novel compression algorithm	Smartphone body shape	Source code of a program

7. International IP Framework

India is a signatory to multiple international IP treaties, which shape and harmonize the domestic statutory regime.

7.1 Paris Convention for the Protection of Industrial Property (1883)

- **Scope:** Patents, designs, trademarks, utility models.
- **Key Principle:** *Right of Priority* – A foreign applicant who files in one member state has a priority date from that first filing, valid for 12 months (patents and utility models) or 6 months (designs) in other member states.
- **National Treatment:** Each member must grant nationals of other members the same protection it grants its own nationals.

7.2 Berne Convention for the Protection of Literary and Artistic Works (1886)

- **Scope:** Copyright and related rights.
- **Key Principle:** *Automatic Protection* – Copyright arises automatically upon creation; no formalities (such as registration) may be imposed.
- **National Treatment and Minimum Standards** (e.g., life + 50 years minimum for literary works).

7.3 TRIPS Agreement (Trade-Related Aspects of Intellectual Property Rights, 1994)

- **Administered by:** World Trade Organization (WTO).
- **Scope:** Comprehensive – covers patents, designs, copyrights, trademarks, geographical indications, trade secrets, integrated circuit layouts.
- **Key Mandates:**
 - Patent term minimum of 20 years from filing date.
 - Computer programs protected as literary works under Berne.
 - Databases protected under the "sui generis" provision.
 - Rental rights for computer programs and cinematographic works.

7.4 Patent Cooperation Treaty (PCT, 1970)

- **Administered by:** World Intellectual Property Organization (WIPO).
- **Function:** A unified procedure for filing patent applications in multiple jurisdictions through a single international application.
- **Phases:**
 1. **International Phase:** Filing and international search (and optional preliminary examination).
 2. **National Phase:** Within 30/31 months from priority date, the applicant enters individual jurisdictions for substantive examination and grant.

India joined the PCT on **7 December 1998**, designated the **Indian Patent Office as a Receiving Office (RO)**, and later acquired status as an **International Searching Authority (ISA)** and **International Preliminary Examining Authority (IPEA)**.

7.5 Hague Agreement Concerning the International Registration of Industrial Designs (1925, revised 1999)

- Provides a mechanism for international registration of industrial designs, designating multiple member states through a single filing.
- India acceded to the Geneva Act of the Hague Agreement on 17 March 2021.

7.6 WIPO Copyright Treaty (WCT) and WIPO Performances and Phonograms Treaty (WPPT) – The "Internet Treaties" (1996)

- Address challenges posed by digital technology.
- Extend protection to the digital dissemination of copyrighted works.
- India signed both treaties but has not yet ratified them.

8. Indian IP Framework: A Snapshot

Form of IP	Statute	Key Sections	Administering Body
Patents	Patents Act, 1970	Sections 2, 3, 4, 11, 48, 53	Controller General of Patents, Designs & Trade Marks (CGPDTM)
Designs	Designs Act, 2000	Sections 2, 4, 11, 22	Patent Office (Design Wing)
Copyrights	Copyright Act, 1957	Sections 2, 13, 14, 22, 44, 57	Copyright Office, Department for Promotion of Industry and Internal Trade (DPIIT)
Trademarks	Trade Marks Act, 1999	Sections 2, 9, 18, 27	Trade Marks Registry

9. Practical Case Study: Software Innovation

Scenario: A team of B.Tech (CS) students develops a novel image compression algorithm (Algorithm X) and produces a full-stack mobile application implementing it. The team wishes to protect their intellectual contribution.

Analysis:

1. **The Algorithm X (as a method):** If the algorithm produces a technical effect (e.g., reduced bandwidth utilization at no measurable loss in perceptual quality), it may qualify as a **patentable computer-implemented invention** under Section 2(1)(j) and Section 3 of the Patents Act, 1970. The team should consider filing a patent application before any public disclosure.
2. **The Source Code:** Automatically protected by **copyright** under Section 13(1)(a) of the Copyright Act, 1957. The team holds copyright in the source code as authors. They may also register the code with the Copyright Office for prima facie evidentiary value.
3. **The Mobile App Icon and UI Layout:** The icon may qualify as an **artistic work** under copyright. The visual appearance of the application's screens, if registered, may qualify as an **industrial design** under the Designs Act, 2000 (post-2008 amendment permitting GUI registration).
4. **The Brand Name:** A separate application under the **Trade Marks Act, 1999** would be appropriate.
5. **The Patent Application under PCT:** If the team intends to seek protection in multiple countries, they may file a single PCT application designating their chosen jurisdictions. The 30/31 month national phase deadline must be tracked.

Strategic Recommendation: A layered IP strategy—combining copyright, patent, and design protection—is often the most robust approach for software innovations.

10. Summary

This lecture has established the foundational taxonomy of intellectual property rights most relevant to computer science research:

1. **Patents** protect new functional inventions for 20 years, require full disclosure, and confer a negative right to exclude others from making, using, or selling the invention. They are governed by the Patents Act, 1970, and the international PCT and Paris Convention systems.
2. **Industrial Designs** protect the visual appearance of articles for up to 15 years, governed by the Designs Act, 2000, and the Hague Agreement. They are relevant to computer science primarily through GUI design and the appearance of computing devices.

3. **Copyrights** protect original expression, including software source code, for the life of the author plus 60 years. They arise automatically and confer both economic and moral rights under the Copyright Act, 1957 and the Berne Convention.
4. The **international framework** — Paris, Berne, TRIPS, PCT, Hague — provides the harmonized architecture within which Indian IP law operates.
5. For computer science researchers, the choice between (or combination of) these regimes depends on the nature of the intellectual contribution: a novel technical method warrants a patent; source code automatically attracts copyright; the visual design of a product or interface warrants design registration.

11. Exam Preparation Tips

When preparing for examinations in Research Methodology, focus on the following:

1. **Memorize the statutory durations** for each form of IP, as these are frequently tested numerical facts:
 - Patent: 20 years from filing date.
 - Design: 10 + 5 = 15 years.
 - Copyright (literary/artistic): Life + 60 years.
 - Copyright (films/sound recordings): 60 years from publication.
2. **Learn the relevant statutory sections** of each Act — they are often asked in short-answer questions (e.g., "What is the duration of copyright under the Copyright Act, 1957?" — answer with reference to Sections 22–28).
3. **Practice comparative questions** using the comparison table — examiners often pose questions like *"Distinguish between patents and copyrights in five points."*
4. **Connect international conventions to national law.** Map each treaty to the corresponding Indian statute (Berne → Copyright Act; Paris → Patents Act; Hague → Designs Act).
5. **Application questions** are common in the higher-order learning levels. Be prepared to analyze a scenario (e.g., a software innovation) and recommend the appropriate IP strategy.
6. **Use precise legal terminology.** The terms "negative right," "national treatment," "priority date," "PCT national phase," and "idea-expression dichotomy" carry specific meaning.

12. Practice Questions

12.1 Short Answer Questions (2–3 marks each)

1. Define "invention" under Section 2(1)(j) of the Patents Act, 1970.
2. What is the term of protection for an industrial design under the Designs Act, 2000?
3. Distinguish between moral rights and economic rights under the Copyright Act, 1957.
4. What is meant by the "idea-expression dichotomy" in copyright law?
5. State the key principle of the Paris Convention.
6. Why is computer software protected as a "literary work" under the Berne Convention?
7. List three categories of works in which copyright subsists under Section 13 of the Copyright Act, 1957.

12.2 Long Answer Questions (5–10 marks each)

1. Compare and contrast **patents and copyrights** under Indian law with reference to subject matter, duration, registration requirements, and rights conferred.
2. Discuss the **patent eligibility of computer-implemented inventions** in India. Refer to Section 3 of the Patents Act, 1970 and relevant case law.
3. Explain the **Patent Cooperation Treaty (PCT)** system. How does the international phase differ from the national phase? Discuss India's role as a Receiving Office.
4. A software startup develops an innovative cloud-based encryption service. Advise the startup on the **appropriate IP protection strategy**, citing relevant sections of the Patents Act, 1970 and the Copyright Act, 1957.
5. Critically analyze the **TRIPS Agreement's** impact on Indian patent law, particularly with respect to software patents and the term of protection.

12.3 Multiple Choice Questions (1 mark each)

1. The term of a patent in India is:
 - (a) 15 years from filing
 - (b) 20 years from filing
 - (c) 10 years from grant
 - (d) Life of inventor + 60 years
2. Moral rights of an author under the Copyright Act, 1957 are:
 - (a) Assignable
 - (b) Perpetual and inalienable
 - (c) Valid only during the author's lifetime
 - (d) Available only to corporate authors
3. Industrial designs protect:
 - (a) Functional features of an article
 - (b) Aesthetic features of an article
 - (c) The underlying idea
 - (d) The manufacturing process
4. The Berne Convention pertains to:
 - (a) Patents
 - (b) Designs
 - (c) Copyrights
 - (d) Trade marks
5. The Hague Agreement (Geneva Act, 1999) is concerned with:
 - (a) International registration of patents
 - (b) International registration of designs
 - (c) International copyright registration
 - (d) International classification of trademarks

Answers: 1-(b), 2-(b), 3-(b), 4-(c), 5-(b).

13. Glossary of Key Terms

- **Antecedent Art / Prior Art:** Information publicly available before the priority date that may affect the novelty or inventive step of an invention.
- **Compulsory License:** A license granted by the government, without the patent holder's consent, under specified conditions (e.g., Section 84 of the Patents Act, 1970).
- **Confidential Information / Trade Secret:** Information that is not generally known, has economic value, and is subject to reasonable steps to maintain secrecy.
- **Derivative Work:** A work based on or adapted from one or more pre-existing works.
- **First-to-File System:** A patent system where rights are awarded to the first party to file the application, regardless of who invented first.
- **Industrial Application:** The requirement that an invention be capable of being made or used in some kind of industry.
- **Inventive Step:** A feature of an invention that, when viewed against the prior art, would not be obvious to a person skilled in the art.
- **Moral Rights:** Non-economic rights of an author to claim authorship and to preserve the integrity of the work.
- **National Treatment:** The principle under which a country treats foreign nationals the same as its own nationals with respect to IP protection.
- **PCT National Phase:** The stage at which an international application under the PCT is converted into individual national or regional patent applications.
- **Priority Date:** The date from which the patent's novelty is assessed, typically the date of first filing of a patent application.
- **Person Skilled in the Art (PSA):** A hypothetical person having ordinary skill and knowledge in the relevant technical field.

- **Right of Priority:** The right to claim an earlier filing date in a Paris Convention member state for purposes of subsequent filings in other member states.
 - **Sui Generis:** A Latin term meaning "of its own kind" — referring to a unique form of protection that does not fit existing categories (e.g., database protection under TRIPS).
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14. Suggested Further Reading

1. **B.L. Wadhera**, *Law Relating to Intellectual Property*, 5th Edition, Universal Law Publishing.
 2. **V.K. Ahuja**, *Law Relating to Intellectual Property Rights*, 3rd Edition, LexisNexis.
 3. **R.S. Praveen**, *Intellectual Property Rights in India: A General Handbook*, Excel Books.
 4. **World Intellectual Property Organization (WIPO)** – Official website: <https://www.wipo.int> for primary documents and statistics.
 5. **Office of the Controller General of Patents, Designs & Trade Marks (CGPDTM)** – Government of India: <https://ipindia.gov.in>.
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End of Lecture 2 Notes